

2. 实编码 GA 求解函数优化问题 MATLAB 代码

func

```
function result=func2(x)
summ=sum(x.^2);
result=summ;
```

GA

```
%%%%%%%%%%实值遗传算法求函数极值%%%%%%%%%%
%%%%%%%%%%初始化%%%%%%%%%%
clear all;                                %清除所有变量
close all;                               %清图
clc;                                     %清屏
D=10;                                    %基因数目
NP=100;                                  %染色体数目
Xs=20;                                  %上限
Xx=-20;                                 %下限
G=1000;                                 %最大遗传代数
f=zeros(D,NP);                          %初始种群赋空间
nf=zeros(D,NP);                         %子种群赋空间
Pc=0.8;                                 %交叉概率
Pm=0.1;                                 %变异概率
f=rand(D,NP)*(Xs-Xx)+Xx;                %随机获得初始种群
%%%%%%%%%%按适应度升序排列%%%%%%%%%%
for np=1:NP
    FIT(np)=func2(f(:,np));
end
[SortFIT,Index]=sort(FIT);
Sortf=f(:,Index);
%%%%%%%%%%遗传算法循环%%%%%%%%%%
for gen=1:G
    %%%%%%%%%%%采用君主方案进行选择交叉操作%%%%%%%%%%
    Emper=Sortf(:,1);                    %君主染色体
    NoPoint=round(D*Pc);                 %每次交叉点的个数
    PoPoint=randi([1 D],NoPoint,NP/2); %交叉基因的位置
    nf=Sortf;
    for i=1:NP/2
        nf(:,2*i-1)=Emper;
        nf(:,2*i)=Sortf(:,2*i);
        for k=1:NoPoint
            nf(PoPoint(k,i),2*i-1)=nf(PoPoint(k,i),2*i);
            nf(PoPoint(k,i),2*i)=Emper(PoPoint(k,i));
        end
    end
end
%%%%%%%%%%变异操作%%%%%%%%%%
```

```

for m=1:NP
    for n=1:D
        r=rand(1,1);
        if r<Pm
            nf(n,m)=rand(1,1)*(Xs-Xx)+Xx;
        end
    end
end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%子种群按适应度升序排列%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for np=1:NP
    NFIT(np)=func2(nf(:,np));
end
[NSortFIT,Index]=sort(NFIT);
NSortf=nf(:,Index);
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%产生新种群%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
f1=[Sortf,NSortf];           %子代和父代合并
FIT1=[SortFIT,NSortFIT];     %子代和父代的适应度值合并
[SortFIT1,Index]=sort(FIT1); %适应度按升序排列
Sortf1=f1(:,Index);          %按适应度排列个体
SortFIT=SortFIT1(1:NP);      %取前 NP 个适应度值
Sortf=Sortf1(:,1:NP);        %取前 NP 个个体
trace(gen)=SortFIT(1);       %历代最优适应度值
end
Bestf=Sortf(:,1);            %最优个体
trace(end)                   %最优值
figure
plot(trace)
xlabel('迭代次数')
ylabel('目标函数值')
title('适应度进化曲线')

```